A morphology-matched forward-readout protocol for SPARC rotation curves

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Abstract

This paper tests whether residual-blind morphology/readout information can choose a rotation-curve readout family before the residuals are inspected. Standard rotation-curve tests usually compare baryonic Newtonian predictions with universal or weakly morphology-dependent correction laws. This paper develops a different, claim-bounded protocol: before scoring a rotation curve, assign a residual-blind morphology/readout state, select the corresponding readout family, freeze source-side scales and amplitudes where possible, and compare the matched family against wrong-family, shuffled-label, Newtonian, MOND-like, RAR, and fixed predecessor-readout controls. The protocol is motivated by three observations that are made self-contained here: external structural disturbance can correlate with residual structure; residual-shape diagnostics are useful but inverse; and a forward test must choose the readout family without using endpoint residuals. Using SPARC rotation curves as the common numerical arena, the accompanying reproducibility code finds preliminary evidence for morphology specificity: a source-native readout-formula preflight beats wrong readout-formula families in 0.886 of the 44 holdout galaxies. For the shuffled-label null, the mean matched-minus-wrong statistic gives $p \simeq 0.002$ in the 1000-shuffle endpoint run, while the beat-fraction statistic is reported separately below. The matched-versus-wrong signal is preserved in a Newtonian-carrier amplitude-freeze stress test, supporting carrier robustness but not baseline superiority. However, the same preflight beats a fixed TPG/v6 predecessor comparator and MOND-like comparators in only 0.477 of holdout galaxies each, so it is not evidence that the proposed framework outperforms existing baselines as a population law. A separate all-SPARC 175-galaxy morphology triage identifies 39 source-freeze candidates, 61 source/kernel precision cases, 18 morphology/subfamily review cases, and 57 control or low-quality cases. Four inspected single-galaxy readout cases also show encouraging matched-family behavior, but remain heterogeneous and caveated. The analysis contains branch-level negative controls and morphology-underdetermined weak negatives, but no source-complete endpoint-grade true negative. The main conclusion is therefore methodological: morphology-matched forward readouts define a falsifiable endpoint protocol, not a validated gravity theory. A practical next step is a residual-blind readout morphology catalogue that freezes source morphology, projection, H I support, vertical structure, and morphology-trajectory/phase observables before endpoint scoring. The result should be read as a reproducible protocol and internal preflight signal, not as evidence for a new gravitational law.

1 Introduction

The SPARC data set provides high-quality rotation curves and baryonic mass models for 175 disk galaxies [8]. It is a standard testbed for Newtonian baryonic predictions, MOND-like dynamics, and empirical radial-acceleration relations [12, 11, 10]. These comparisons are usually organized around scalar acceleration laws or globally fitted mass-to-light choices. They are powerful precisely because they are simple.

The comparison set in this paper has two different kinds of baselines. The Newtonian, MOND-like, and RAR-like references are external literature-facing comparators. The TPG/v6 curve, by contrast, is an internal predecessor readout/closure comparator from the same research programme. It is included because it is frozen in the reproducibility package and is often a strong smooth baseline. It is not treated here as an independently published reference model, and it is not used to choose or tune the morphology-matched readout family.

The present paper asks a different question. Suppose that some residual structure is not described well by a universal scalar correction, but depends on which morphology/readout state the galaxy occupies: regular disk, extended scale-tail disk, compact finite-source system, vertical/projection-dominated system, ring/resonance structure, warp/trajectory structure, or a mixture of such components. Then the relevant test is not merely whether a flexible curve can fit a galaxy. The test is whether the morphology-matched readout family, chosen before endpoint scoring, performs better than wrong morphology families and shuffled labels, and only then whether it is competitive with conventional baselines.

This paper makes that test self-contained. The framework is used only through operational definitions: a galaxy has a projected morphology handle K_{obs}, a possibly different readout-relevant proxy K_{readout}, and a family of local readout kernels that generate a forward correction $\delta v_K^2(R)$ or equivalent closure response. No external theory paper is required for the protocol. The definitions, allowed inputs, forbidden inputs, endpoint score, negative-result taxonomy, and numerical status are stated below.

1.1 Reader's roadmap

What must be accepted to read the paper: only the operational protocol defined here. A reader does not need to accept a completed new gravity theory.
What is tested: whether a residual-blind morphology/readout label can choose a restricted readout family that performs better than wrong families and shuffled labels.
What is not claimed: population-level superiority over Newtonian, MOND/RAR-like, or TPG/v6-like baselines; a final morphology catalogue; or a validated gravitational theory.
Where the evidence appears: Section 3 defines the scoring protocol; Section 4 defines the readout families; Section 5 gives the reproducibility and leakage-prevention path; Section 6 reports population preflights, the 175-galaxy triage, and four inspected rotation-curve cases; Section 7 explains how to read failures.

1.2 Status of the underlying theory language

The terminology "readout", "closure", and "morphology state" is motivated by an unpublished broader theory programme. That programme is not assumed as a published premise of this manuscript. The present paper therefore uses only the operational layer defined here: residual-blind morphology labels, predeclared readout families, frozen source-side inputs, endpoint scores, wrong-family controls, shuffled-label controls, and conventional comparators. Any reader can evaluate the claims below without accepting the unpublished theory motivation.

1.3 Theory motivation isolated as a testable consequence

The broader motivation can be summarized without importing the full derivation. In the underlying readout picture, the morphology seen in 4D is not assumed to be the complete state that controls the effective gravitational response. It is a projected handle on a deeper readout configuration. Consequently, a galaxy's current visual class K_{obs} may differ from the readout-relevant class K_{readout}, especially when projection, H I support, vertical structure, warp/trajectory evidence, rings, bars, or lopsidedness are present.

In the companion bridge language, this deeper readout configuration is called a *Tau morphology state*. The name does not mean a visible galaxy shape. It denotes a Tau-side organizing state whose four-dimensional readouts may include visible morphology, matter distribution, gravity response, time/clock structure, observer/path appearance, and quantum/coherence structure. The present paper uses only the operational consequence of that idea: the observed 4D morphology label is a source handle, not automatically the active readout-family label.

This distinction is also how the observer/projection intuition enters the paper. The intrinsic Tau-side organization of a galaxy is not treated as a mere observer-dependent illusion. However, the 4D morphology and dynamics available to an observer are projected readouts of that organization. Thus the working hierarchy is

$$K^{\tau}_{\mathrm{intrinsic}} \longrightarrow K^{4D}_{\mathrm{projected}}(\mathcal{O}) \longrightarrow K_{\mathrm{readout}},$$

where $\mathcal{O}$ denotes the observer/path/projection channel. A fuller projection-enriched version also allows a source-frozen morphology trajectory/phase variable Θ_{morph}, representing the possibility that 4D past, present, and future morphology are different readout slices of one deeper Tau-side configuration. The present paper does not model that full observer-time response. It uses this hierarchy only to justify why the observed morphology label should not automatically be identified with the readout family tested against the rotation curve.

The operational consequence is the only part tested here:

$$K_{\mathrm{obs}} \longrightarrow K_{\mathrm{readout}} \longrightarrow \mathcal{F}_{K_{\mathrm{readout}}} \longrightarrow \delta v^2_{K_{\mathrm{readout}}}(R).$$

If this readout interpretation carries empirical information, then a residual-blind morphology-matched family should perform better than wrong morphology families and shuffled labels. This is a weaker and safer statement than claiming a completed gravity theory. The full derivation may motivate the choice of language, but the present paper tests only this forward-readout consequence.

1.4 Mini-glossary

Table 1: Plain-language glossary for the main protocol terms.

Symbol or term	Meaning in this paper
K_{obs}	The residual-blind morphology handle suggested by observed structure: disk, warp, ring, bar, compact support, tail, or lopsidedness.
K_{readout}	The morphology/readout state used to choose the formula family. It may differ from K_{obs} when projection, H I support, vertical structure, or trajectory/phase evidence matters.
Matched family	The readout family selected before endpoint scoring.
Wrong family	A deliberately mismatched family used as a morphology-specific control.
Shuffled labels	A null test where morphology labels are randomized.
TPG/v6	A frozen internal predecessor comparator, not an external published baseline and not a tuning input.
Endpoint	A scored test after the readout family and source inputs have been frozen.
True negative	A source-complete, residual-blind endpoint failure against the relevant baselines and controls.

2 Operational framework

2.1 Observed morphology versus readout morphology

Let K_{obs} denote a residual-blind observed morphology handle: for example thin disk, thick disk, warp, ring, bar, compact support, extended H I tail, or lopsidedness. The quantity used to choose a readout family is K_{readout}. It may equal K_{obs}, but it need not. A present 4D visual class can be only a projection of the readout-relevant state; edge-on projection, vertical structure, H I support, or morphology-trajectory evidence can change which readout family should be tested.

In this sense, the morphology component is partly observer/projection dependent, but not because the galaxy's source structure is assumed to be arbitrary. The source-side organization is the target; the observer sees a 4D projection of it. Inclination, line-of-sight geometry, waveband, H I extent, vertical support, and trajectory/phase indicators can change the available projected proxy. The protocol therefore treats K_{obs} as a residual-blind observational handle and K_{readout} as the source-disciplined class used for forward scoring. The current kernels are projection-aware morphology proxies, not full observer-projection Tau Core readouts; observer/path/time-projection effects enter only through frozen source proxies such as inclination, vertical support, H I extent, warp/trajectory flags, and projected readout-family selection. The kinds of source fields needed for this distinction are standard observational products rather than exotic new observables: rotation-curve and mass-model catalogues [8], H I synthesis and velocity-field studies [22, 24], photometric/kinematic late-type galaxy studies [20], and lopsidedness or asymmetry catalogues [23].

For each family K, let $\mathcal{F}_K$ be a source-first readout shell and let $K_{gK}(R)$ be the radial kernel assigned to galaxy g. The generic velocity-squared form used in this paper is

$$\delta v_{gK}^2(R) = A_{gK} K_{gK}(R),$$

where the amplitude A_{gK} must be either source-derived or frozen by a predeclared rule. The rotation curve v_{obs} may be read only by the separate scoring step.

2.2 No-curve-fitting rule

The construction ladder is

$$\begin{aligned} K_{\text{obs}} &\longrightarrow K_{\text{readout}} \\ &\longrightarrow \{O_{gK}\}_{\text{source}} \\ &\longrightarrow \mathcal{F}_K,\ K_{gK}(R) \\ &\longrightarrow A_{gK}^{\text{source}} \text{ or predeclared amplitude rule} \\ &\longrightarrow \delta v_{gK}^2(R). \end{aligned}$$

Allowed inputs are residual-blind morphology and source quantities: disk scale, H I radius, outer-tail onset, compact support radius, vertical thickness, projection state, bar/ring/lopsidedness evidence, and environment or trajectory/phase context. Forbidden construction inputs are the observed endpoint residual, the best baseline score, inverse diagnostics that ask what preparation would be needed to absorb the residual, and post-hoc family selection.

The common bridge object behind this ladder is not a fitted rotation curve. It is the conditional Tau-side quotient residual

$$\delta g_{\text{morph}}(x) = R_{g,x}^{4D}\left[\rho_D(\mathcal{M}_\tau(x))(Q_D^\tau(x))\right],$$

where Q_D^τ is the ordered-disk readout direction, $\mathcal{M}_\tau$ is the local readout/closure mismatch generator, $\rho_D = \pi_B \circ \mathcal{M}_\tau|_{D_{\mathrm{diag}}}$ projects away the Newtonian, ordinary-disk, and gauge sectors, and R_g^{4D} is the weak-field 4D readout map. Appendix A derives the operational kernels by reducing this same bridge equation to source-frozen one-dimensional radial readouts. This is why the families below are treated as bridge specializations rather than as independent empirical templates.

2.3 Readout family registry

The current first-pass registry is deliberately small and test-oriented. It is not asserted to be a final physical taxonomy.

Table 2: Claim-bounded morphology/readout families used in the present tests.

Family	Morphology/readout handle	Representative shell	Status
K_{tail}	scale-tail or diffuse LSB disk	$\delta v^2 = \xi_{\mathrm{tail}}\Lambda_{\mathrm{tail}} I_2(R)/R$	1D testable
K_{exp}	regular exponential disk	$\delta v^2 = A_{\mathrm{exp}} y^2[I_0K_0 - I_1K_1]$, $y = R/(2R_s)$	1D testable
K_{compact}	compact finite support	exterior finite-source response, $\delta g_R \sim -I_\infty/R^2$	1D testable
K_{flare}	thick/flared or projection-sensitive disk	$v^2_{\mathrm{carrier}}[1 - \Gamma_{\mathrm{vp}} K_{\mathrm{vp}}(R)]$	1D proxy
K_{ring}	ring or resonance structure	localized radial kernel near R_{ring}	1D proxy
$K_{m=2}$	barred spiral	non-axisymmetric harmonic response	velocity-field preferred
$K_{m=1}$	lopsided disk	non-axisymmetric harmonic response	velocity-field preferred

The entries in Table 1 are not arbitrary curve templates. They are the minimal source-first kernels retained after the readout motivation is reduced to one-dimensional rotation-curve observables. The reduction has three steps:

$$\text{morphology/readout handle} \longrightarrow \text{allowed radial source scale} \longrightarrow \text{restricted kernel shape.}$$

For example, a scale-tail or diffuse LSB handle permits an extended outer-load kernel; an exponential-disk handle permits the standard finite exponential-disk radial carrier shape; a compact-support handle permits an exterior finite-load falloff; and a thick/flared or projection-sensitive handle permits a vertical/projection attenuation kernel. Ring, bar, and lopsided handles are listed because they are part of the morphology registry, but in the present one-dimensional SPARC analysis they remain proxy or velocity-field-preferred families rather than full endpoint kernels.

Table 3: Current sample coverage of the source-native readout families. Counts refer to the 175-row SPARC preflight label table. The examples are illustrative, not a complete list.

Family	Total	Holdout	Example assigned galaxies	Kernel origin in this paper
K_{tail} / $K_{\text{scale\_tail\_spiral}}$	80	20	CamB, F561-1, F563-1, IC2574, NGC4214, NGC6789	outer-load / scale-tail source handle reduced to a radial tail kernel
K_{exp} / $K_{\text{exponential\_disk}}$	32	7	ESO116-G012, NGC4183, UGC02259, UGC07151, UGC07261, UGC07603	regular disk handle reduced to a finite exponential-disk carrier shape
K_{compact} / $K_{\text{compact\_finite}}$	29	7	IC4202, NGC4013, NGC5985, NGC6946, UGC03580, UGC08699	compact finite-source handle reduced to an exterior finite-load response
K_{flare} / $K_{\text{thick\_flared}}$	34	10	F568-1, NGC0024, NGC2683, NGC3726, NGC3949, NGC3972	vertical, edge-on, or projection-sensitive handle reduced to an attenuation proxy

The current population scoring uses the four one-dimensional families in Table 3. The registry also records ring, barred, and lopsided families, but those require either stronger source fields or two-dimensional velocity-field information before they should be promoted to the same endpoint status. This is why the paper's numerical claims are stated for the source-native 1D preflight families and for four inspected single-galaxy cases, not for the full conceptual registry. Appendix A gives the common bridge equation and the explicit readout-kernel derivation ledger for the morphology families used here, including the dimensional status and the precise claim boundary for each family.

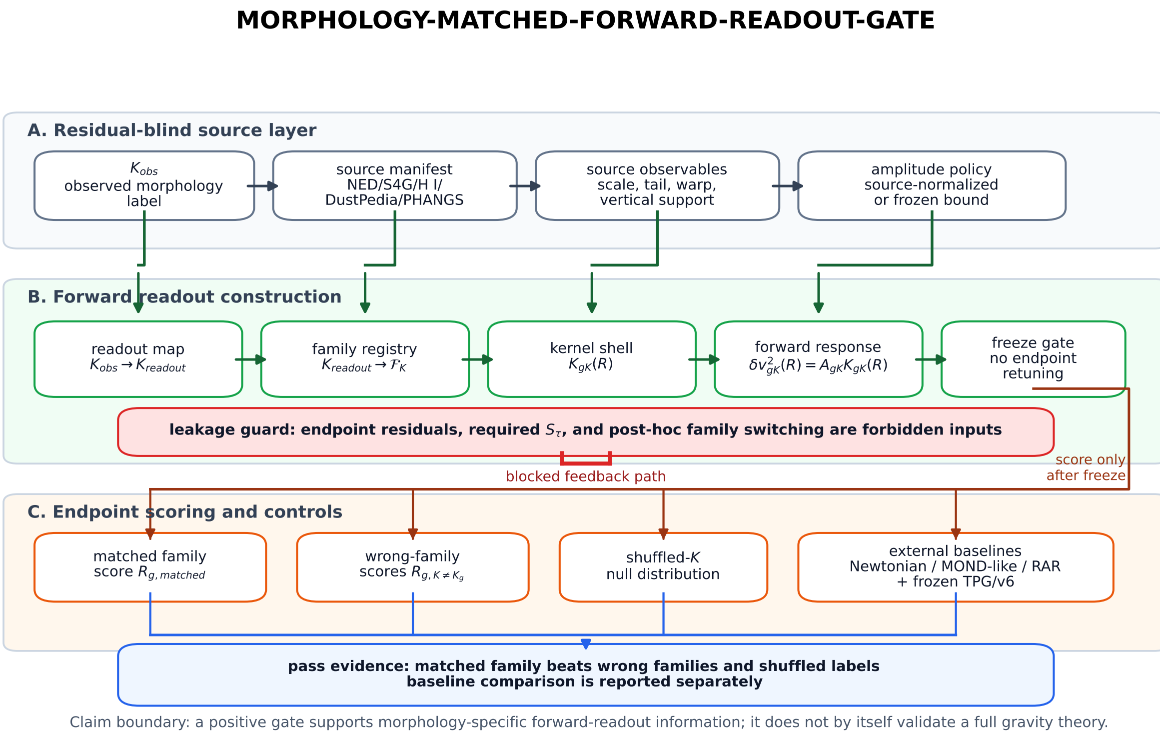

Figure 1: Morphology-matched forward-readout gate. Residual-blind source fields first determine the observed morphology handle, readout-relevant proxy, source observables, and amplitude policy. Only after the readout family, kernel shell, forward response, and freeze gate are fixed may endpoint scores be computed against matched, wrong-family, shuffled-label, and baseline controls. The red guard marks forbidden leakage from endpoint residuals or required-S_τ diagnostics back into family selection.

3 Endpoint protocol

For galaxy g, define $R_{g,K}$ as the residual score after applying family K. In the present numerical work this is an RMSE in rotation speed. The matched score is

$$R_{g,\mathrm{matched}} = R_{g,K_{\mathrm{readout},g}},$$

and the wrong-family mean is

$$R_{g,\mathrm{wrong}} = \frac{1}{|\mathcal{K}|-1} \sum_{K \neq K_{\mathrm{readout},g}} R_{g,K}.$$

The morphology-specific endpoint is

$$\Delta_{\mathrm{matched}} = \mathrm{RMS}_g(R_{g,\mathrm{wrong}}) - \mathrm{RMS}_g(R_{g,\mathrm{matched}}).$$

The matched family should also rank well among all candidate families:

$$\mathrm{rank}_g(K_{\mathrm{readout},g}) = \mathrm{rank}\left(R_{g,K_{\mathrm{readout},g}} \text{ among } \{R_{g,K} : K \in \mathcal{K}\}\right).$$

A positive result requires comparison not only to wrong families, but also to shuffled morphology labels and conventional baselines. A morphology-specific win against wrong families is therefore weaker than a full all-baseline population win.

4 Data, controls, and scoring protocol

The numerical arena is SPARC [8]. The current scripts use the available SPARC rotation-curve points and the baseline columns already present in the reproducibility package: Newtonian baryonic, TPG/v6, and a MOND-like comparator. The analysis is organized into a train/holdout split with 131 train galaxies and 44 holdout galaxies for the population preflights. Where population amplitudes are needed, they are estimated on the train split only and then evaluated on the holdout split. This is weaker than a fully source-frozen amplitude theorem, but stronger than fitting each holdout galaxy independently.

The controls have separate roles:

- *Wrong-family controls* test morphology specificity within the readout family registry.
- *Shuffled-label controls* test whether the morphology labels carry information beyond generic flexibility.
- *Newtonian, TPG/v6, and MOND-like controls* test whether the selected readout is competitive with conventional or already-available comparators.
- *Single-galaxy endpoint controls* test whether an object-specific source-frozen protocol is internally nonleaky before any broader claim is made.

4.1 Status of the TPG/v6 comparator

The TPG/v6 column requires a separate caveat. It is not a standard external literature baseline in the same sense as Newtonian, MOND-like, or RAR-like comparisons. It is a fixed predecessor readout from the author's earlier closure-oriented programme. In this manuscript it is used operationally: a galaxy either beats the frozen TPG/v6 curve or it does not. The endpoint code does not rederive TPG/v6, does not retune it per galaxy, and does not use it to select K_{readout}.

TPG/v6 use in this paper: frozen before the endpoint runs; not used for label selection; not retuned per galaxy; included only as an internal predecessor-control; not claimed as an external literature baseline.

This distinction matters for interpretation. If a morphology-matched readout beats wrong families but not TPG/v6, the result still supports morphology specificity but not baseline superiority. If it beats TPG/v6 in a source-frozen endpoint, that is evidence against a strong predecessor-control curve, not evidence against an independently established external theory.

4.2 Score definitions

The main score is RMSE in rotation speed, quoted in $\mathrm{km\,s^{-1}}$. Lower scores are better. For difference scores such as matched-minus-wrong, negative values mean that the matched readout improves on the comparator. The permutation nulls use 1000 deterministic shuffles in the current executable analysis. The reported p-values are therefore preparation-level finite-shuffle estimates, not asymptotic discovery claims.

For a galaxy g, model m, and measured radii R_i, the plotted and tabulated endpoint score is

$$\mathrm{RMSE}_{g,m} = \left[\frac{1}{N_g}\sum_{i=1}^{N_g}\left(v_m(R_i) - v_{\mathrm{obs}}(R_i)\right)^2\right]^{1/2}.$$

The endpoint residuals shown below are

$$\Delta v_m(R_i) = v_m(R_i) - v_{\rm obs}(R_i).$$

Thus a visually smooth curve is not sufficient; it must reduce the signed residual structure without having been selected from those residuals.

5 Reproducibility and leakage prevention

The numerical claim in this manuscript is meaningful only if the morphology labels, family choices, and source scales are fixed before endpoint residuals are scored. The reproducibility package therefore separates source artifacts, amplitude artifacts, scoring artifacts, and paper-generation artifacts.

5.1 Exact data sources and frozen artifacts

The released package uses relative repository paths. One upstream SPARC/TPG point table was read from the author's local run directory while preparing this draft; in a release archive it should be copied or symlinked into the package under a relative path before reproduction. The exact inputs used by the source-native readout-formula preflight are:

- *Rotation curves and baseline columns*: the SPARC-based point table `data/external/tau_rotation_curve_frozen_proxy_runner_v0_points.csv` in the released package. In the author's local run this was read from the TPG results directory. This table supplies R, $v_{\rm obs}$, σ_v, Newtonian/proxy columns, TPG/v6, and MOND-like comparator columns used by the executable preflight.
- *SPARC metadata*: `data/derived/external_sparc_master_table.csv`, derived from the SPARC master table [8]. This supplies galaxy type, distance, inclination, luminosity, disk scale, surface brightness, H I mass/radius where available, and quality flags.
- *Residual-blind morphology parameter manifest*: `data/derived/morphology_parameter_manifest.csv`. This is the source manifest consumed by the formula endpoint script.
- *Label-freeze artifact*: `data/derived/source_native_readout_formula_labels.csv`. It records the train/holdout split, formula family, manifest confidence, source proxies, amplitude policy, parameter source, caveats, and forbidden fields for all 175 SPARC galaxies.
- *Amplitude-freeze artifact*: `data/derived/source_native_readout_formula_amplitudes.csv`. It records one family-level amplitude per readout family, estimated on train galaxies only.

The population split used in the preflight is the frozen split column consumed from the point/label artifacts: 131 train galaxies and 44 holdout galaxies. The endpoint scripts do not redraw this split for the headline result. The alternative-split robustness audit reported below is a separate stress test, not the primary endpoint.

5.2 Amplitude freeze rule and forbidden endpoint fields

For the source-native formula preflight, the amplitude rule is train-only and family-level:

$$A_K = \arg\min_A \sum_{g\in{\rm train},\, K_g=K} \sum_i \left[v^2_{\rm obs}(R_{gi}) - v^2_{\rm TPG/v6}(R_{gi}) - A K_{gK}(R_{gi}) \right]^2.$$

The fitted A_K is then applied unchanged to holdout galaxies. This is not a source-derived amplitude theorem; it is a preparation-level amplitude freeze. The TPG/v6 curve is used only as the frozen carrier for this amplitude-residual preflight. It is not used to select labels or readout families, and holdout scoring still reports matched-versus-wrong, TPG/v6, and MOND-like comparisons separately. The paper therefore treats the population result as evidence for morphology-specific formula choice, not as final empirical validation.

Forbidden fields for label and family construction are:

$$v_{\mathrm{obs}}, \quad v_m - v_{\mathrm{obs}}, \quad \text{RMSE}, \quad \text{best baseline score}, \quad \text{required } S_\tau, \quad \text{post-hoc family choice}.$$

The label-freeze artifact records this explicitly in its `forbidden_inputs` column. Endpoint residuals may be used only after K_{readout}, source proxies, kernels, and amplitude policy have been written to disk.

5.3 Scripts, seeds, and outputs

The shortest path to reproduce the headline source-native result is:

```
python scripts/reproduce.py
python scripts/run_source_native_readout_formula_endpoint.py
python scripts/build_source_native_readout_robustness.py
python scripts/run_source_native_carrier_robustness.py
cat data/derived/source_native_readout_formula_endpoint_summary.csv
```

The corresponding robustness table is written to `data/derived/source_native_readout_formula_robustness_summary.csv`; the carrier stress test is written to `data/derived/source_native_carrier_robustness_summary.csv`.

The executable path is:

1. `scripts/build_morphology_parameter_manifest.py`: builds the residual-blind morphology/source manifest.

2. `scripts/run_source_native_readout_formula_endpoint.py`: builds readout kernels, fits train-only family amplitudes, scores train and holdout, and runs the 1000-shuffle null.

3. `scripts/build_source_native_readout_robustness.py`: runs the 10k shuffle, bootstrap, error-weighted, alternative-split, and family-balanced robustness checks.

4. `scripts/run_source_native_carrier_robustness.py`: repeats the train-only family-amplitude freeze with TPG/v6 and Newtonian baryonic carriers while keeping labels, kernels, source proxies, and splits fixed.

5. `scripts/build_paper_summary_figures.py` and `scripts/generate_paper8_artifacts.py`: regenerate paper figures and protocol diagrams from saved CSV artifacts.

6. `scripts/reproduce.py`: repository-level reproduction driver.

7. `scripts/build_arxiv_source.py`: builds the arXiv source package.

The random seeds are fixed: 31415 for the original 1000 shuffled-label null, 271828 for the 10k shuffled-label robustness null, 271829 for the bootstrap confidence interval, and 271830 for the alternative-split robustness audit.

The output artifacts used by the paper are:

- data/derived/source_native_readout_formula_scores_by_galaxy.csv and data/derived/source_native_readout_formula_endpoint_summary.csv;
- data/derived/source_native_readout_formula_shuffled_null.csv and data/derived/source_native_readout_formula_shuffled_null_summary.csv;
- data/derived/source_native_readout_formula_robustness_summary.csv and data/derived/source_native_readout_formula_error_weighted_scores.csv;
- data/derived/source_native_carrier_robustness_summary.csv, data/derived/source_native_carrier_robustness_scores_by_galaxy.csv, data/derived/source_native_carrier_robustness_by_family.csv, data/derived/source_native_carrier_robustness_freeze_invariance_audit.csv, and data/derived/source_native_carrier_robustness_shuffled_null_summary.csv;
- data/derived/all_galaxy_morphology_precision_triage.csv;
- data/derived/four_case_endpoint_status_cases.csv.

Leakage can be checked by verifying that the label artifact contains no endpoint residual or score columns, that its parameter_source fields are source/proxy descriptors, that the forbidden_inputs column lists endpoint residuals and required-S_τ diagnostics as forbidden, and that the amplitude artifact contains only family-level train-only amplitudes. The holdout scores in source_native_readout_formula_scores_by_galaxy.csv are produced only after those artifacts exist.

5.4 All-galaxy morphology triage

In addition to the 44-galaxy holdout preflights and the four inspected endpoints, the reproducibility package contains an all-SPARC morphology precision triage over the full 175-galaxy SPARC table. This step is not an endpoint validation. It is a worklist compiler: for every galaxy it assigns a current readout family, records which source observables are missing, scores the currently compiled proxy against available rotation data, and classifies the case by what should happen next.

The important methodological point is that this triage is deliberately one-way. The readout family and missing-source diagnosis are not promoted from endpoint residuals. Endpoint velocities are used only after the current compiled protocol exists, in order to decide whether the case is already useful, needs source/kernel precision, needs morphology or subfamily review, or should be kept as a low-quality/control object. The resulting classes are summarized in Table 4.

Table 4: All-SPARC morphology precision triage. These 175 rows are a source-acquisition and protocol-prioritization product, not endpoint validation.

Triage class	Galaxies	Proxy baseline flag fraction	Interpretation
Current morphology good enough for source-freeze review	39	1.000	Promising candidates for predeclared replay or source-frozen endpoint work
Morphology signal present, needs source/kernel precision	61	0.000	Matched family often beats wrong families, but not the best baseline yet
Morphology or subfamily review needed	18	0.000	Likely readout-family, subfamily, or kernel mismatch; not a true negative
Control or low quality, do not prioritize	57	0.281	Useful mainly as controls or after rotation/source-quality improvement

This table is the paper's broadest survey layer. It says that the current readout proxies carry morphology information on many galaxies, but also that the limiting factor is often source completeness and readout-subfamily precision rather than the existence of a fitted curve. In particular, the 39 "good enough" cases are not claimed as accepted endpoints; they are the highest-priority queue for residual-blind source freezing. The 18 morphology/subfamily-review cases are also not failures of the framework; they are the clearest places where the present readout label may be too coarse.

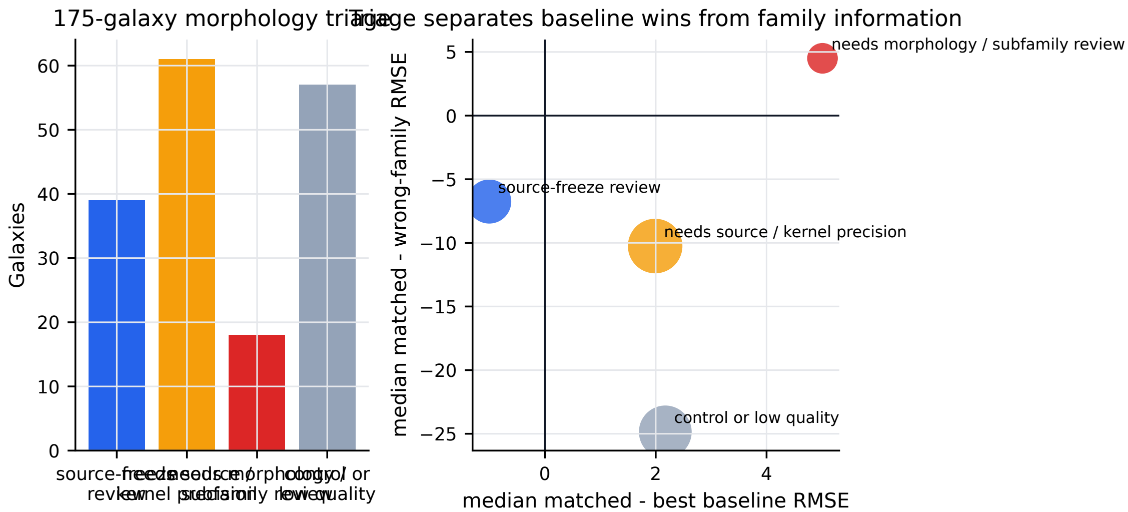

Figure 2: All-SPARC morphology precision triage. Left: the 175 galaxies are partitioned into source-freeze candidates, source/kernel precision cases, morphology/subfamily review cases, and control/low-quality cases. Right: median score differences show why the classes should not be collapsed into a single win/loss statistic. Some classes carry strong wrong-family information without yet beating the best baseline.

5.5 How morphology labels were assigned

The morphology labels used in the present paper have different evidence grades. This is why the manuscript distinguishes population preflights, all-galaxy triage, and inspected single-galaxy endpoints. The broad population layers use available residual-blind metadata and proxy source fields; the four inspected cases use object-specific source ledgers. Table 5 summarizes the assignment logic.

Table 5: Morphology/readout source audit. "Proxy" means sufficient for preflight or triage but not for endpoint validation.

Layer or galaxy	Readout label status	Source basis	Claim boundary
44-galaxy holdout preflights	proxy family labels	SPARC rotation/mass-model metadata, type, inclination, radial coverage, surface-brightness and gas/bulge proxy fields; train-only amplitude policies	Preparation-level screen only; useful for matched-vs-wrong and shuffled-label tests, not endpoint validation
175-galaxy all-SPARC triage	compiled current readout family	Accepted-manifest table plus missing-observable ledger: source status, confidence, caveats, missing fields, fallback policy, and next source gate	Worklist and prioritization layer; endpoint velocities are used only after the compiled protocol exists
NGC4013	mixed exponential-disk / warp-vertical overlay lane	H I and vertical-structure literature context [27, 3]; later fresh-source review supports the lane, but the score remains retrospective	Protocol-ready and replay-needed; not an accepted population endpoint
NGC5907	projection-dominated mixed readout	Warp, edge-on, and interaction/projection context [16, 18, 26]	Accepted preliminary single-galaxy control; population validation still open
NGC7331	exponential/vertical/outer-warp mixed lane	H I warp/trajectory and vertical-scale context [2, 14]; pure branch retained as negative control	Caveated accepted mixed case; broad outer window and pure-branch negative remain explicit caveats
NGC4088	warp/trajectory readout	H I/interaction and warp/asymmetry source packet from the object-specific ledger; B1 provenance and B2/B3 normalization remain caveated	Strongest single inspected case; caveated endpoint, not a general law claim

Thus the morphology determination is not uniform across all rows. The broad screens deliberately use weaker but residual-blind proxies to find where the protocol is informative. The inspected galaxies use stronger source packets, but they still carry case-specific caveats. This is why the paper does not collapse the evidence into a single "Tau wins" or "Tau fails" statement.

6 Numerical evidence

Claim levels used below.

- **Strong internal-preflight signal:** source-native readout formulas beat wrong morphology families on the holdout split more often than shuffled labels would predict.
- **Moderate case evidence:** four inspected galaxies show local matched-family improvement, but each remains caveated.
- **Not shown:** population-level superiority over Newtonian, MOND/RAR-like, or TPG/v6-like baselines.

6.1 Population preflights

The current executable analysis contains several preparation-level screens. They are useful because they separate morphology specificity from baseline superiority.

Table 6: Main population-level preflight outcomes. These are internal preflight results, not empirical validation.

Test	Holdout/sample	Beats wrong family	Beats TPG/v6	Beats MOND-like
Metadata readout proxy	44 holdout	0.568	0.636	0.614
Naive radial shell proxy	44 holdout	0.500	0.477	0.477
Source-native readout formulas	44 holdout	0.886	0.477	0.477
Readout-mixture proxy	44 holdout	0.455	0.500	0.455
L1 source-light screen	66 screened	53/66	17/66 all-baseline	–

The strongest internal-preflight signal is the source-native readout-formula preflight: the matched formula beats the wrong-formula mean in 0.886 of the 44 holdout galaxies, with a 1000-permutation shuffled-label null $p \simeq 0.002$ for the mean matched-minus-wrong statistic and $p \simeq 0.006$ for the beat-fraction statistic. This supports the morphology-family specificity hypothesis. It does not support a stronger claim that the present readout formulas beat TPG/v6, MOND-like, or RAR-like baselines as a population law.

Table 7 records robustness checks for the same source-native formula result. These checks are not independent discoveries; they are stress tests of the headline matched-versus-wrong statistic.

Table 7: Robustness checks for the 0.886 holdout matched-versus-wrong fraction. All rows use the frozen source-native formula labels and are generated by `scripts/build_source_native_readout_robustness.py` or `scripts/run_source_native_carrier_robustness.py`.

Test	Result	p-value	Detail
Original holdout	0.886	–	44 holdout galaxies; mean matched-minus-wrong $= -17.684\,\mathrm{km\,s^{-1}}$.
10k shuffled-label null	0.886	0.0093	Seed 271828; beat-fraction statistic; null beats-wrong median = 0.773.
Bootstrap CI	0.886	–	Seed 271829; 10000 galaxy bootstrap interval $[0.795, 0.977]$.
Error-weighted RMSE	0.886	–	Uses SPARC velocity-error weights; mean weighted matched-minus-wrong $= -17.758\,\mathrm{km\,s^{-1}}$.
Alternative split median	0.841	–	Seed 271830; 100 family-stratified splits; central 90% range $[0.773, 0.909]$.
Family-balanced score	0.832	–	Mean of holdout per-family beat fractions; compact 0.571, exponential 0.857, scale-tail 1.000, thick/flared 0.900.
Newtonian-carrier amplitude freeze	0.955	0.0010	Train-only amplitudes fit around v_n^2 rather than TPG/v6; mean matched-minus-wrong $= -51.131\,\mathrm{km\,s^{-1}}$, median $-53.425\,\mathrm{km\,s^{-1}}$; family-balanced beat fraction = 0.929; beat-fraction null $p = 0.0010$.

The Newtonian-carrier row is important because it asks whether morphology specificity survives when the train-only amplitude residual is defined around the baryonic Newtonian carrier v_n rather than the TPG/v6 carrier. It does: the matched-versus-wrong holdout fraction remains high, the median margin has the same sign as the mean, and the family-balanced score remains high. The freeze-invariance audit in `data/derived/source_native_carrier_robustness_freeze_invariance_audit.csv` records that labels, kernels, source proxies, and train/holdout splits are unchanged. This strengthens the carrier-robustness of the morphology-specific signal, but it still does not establish population-level superiority over TPG/v6, MOND-like, RAR-like, or Newtonian baselines.

This outcome is not treated as a surprise to be hidden. It is part of the protocol. If a galaxy is morphologically regular, approximately scalar in its effective residual structure, or dominated by a smooth closure-like response, then Newtonian, MOND/RAR-like, or TPG/v6-like curves can be strong. In the language of the present paper, those baselines may be good low-dimensional readout limits. The morphology-matched programme becomes distinctive only where the source morphology selects a different readout subfamily and that choice improves against wrong families and shuffled labels before any endpoint repair.

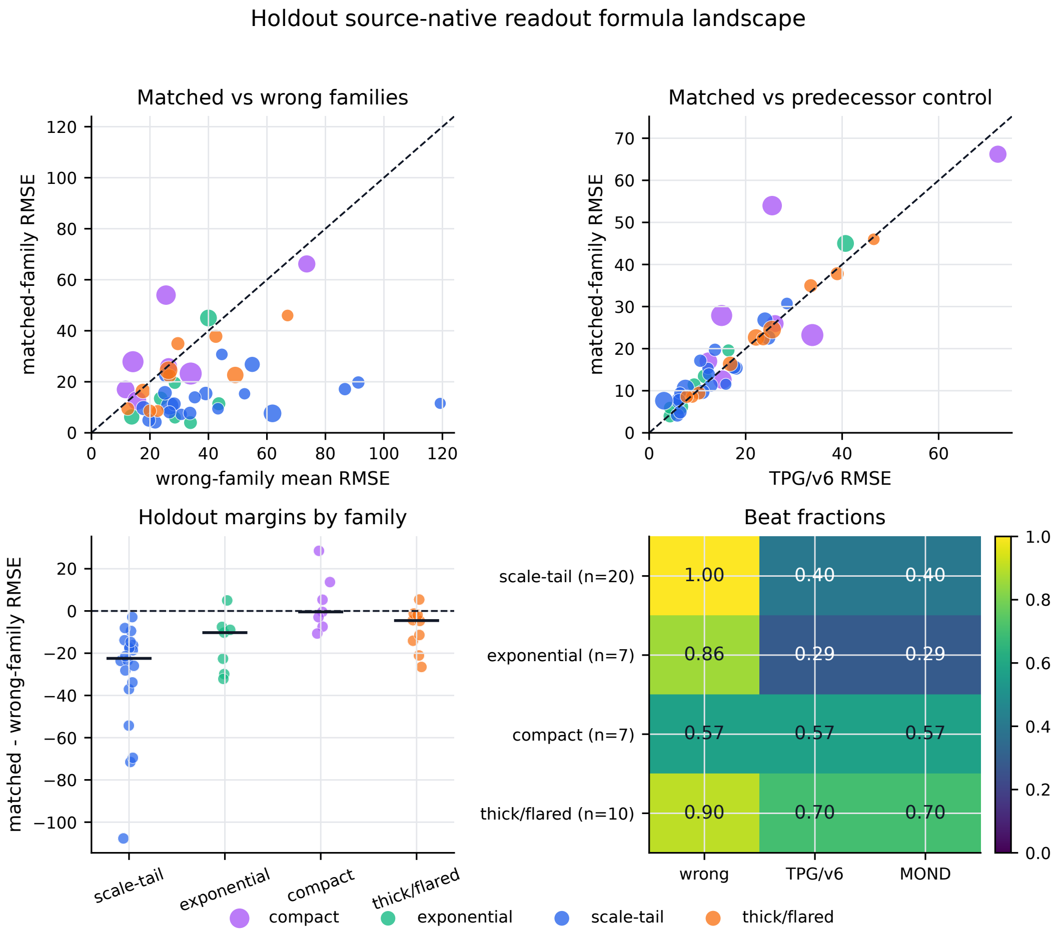

Figure 3: Holdout source-native readout formula landscape. The upper panels compare matched-family RMSE against wrong-family and TPG/v6 controls for each holdout galaxy. Marker color denotes readout family and marker size tracks the number of rotation-curve points. The lower panels show family-level margins and beat fractions. The figure makes the main result visual: the matched families often beat wrong morphology families, while baseline superiority remains mixed.

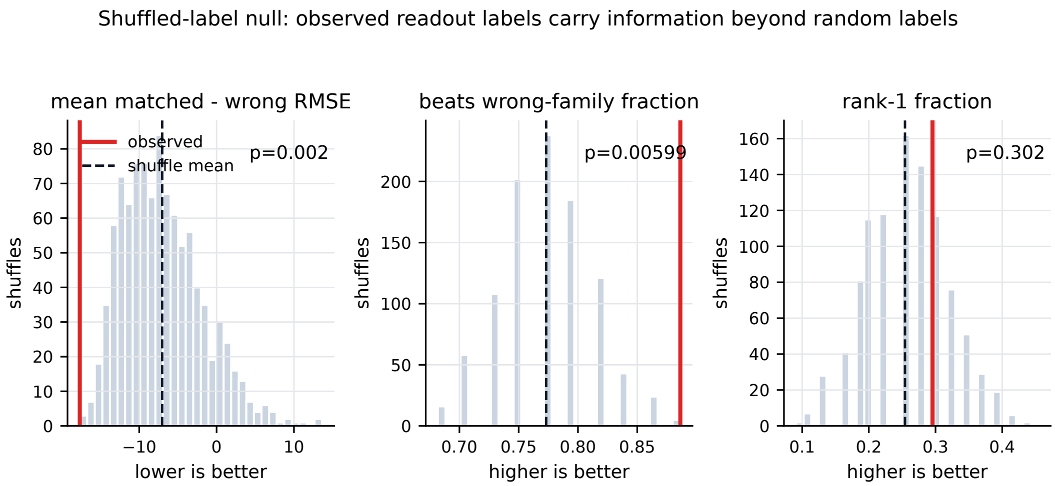

Figure 4: Shuffled-label null distributions for the source-native readout formulas. Grey histograms show 1000 morphology-label shuffles; red lines show the observed holdout statistic. The observed matched-minus-wrong mean and beats-wrong fraction lie in the favorable tail of the shuffled null, while the rank-1 statistic is weaker.

The negative readout-mixture proxy is also informative. It shows that a mixture layer cannot be supplied by coarse present-day proxy heuristics. Mixture weights need accepted source observables: projection, H I support, vertical structure, interaction/history context, and a residual-blind normalization rule.

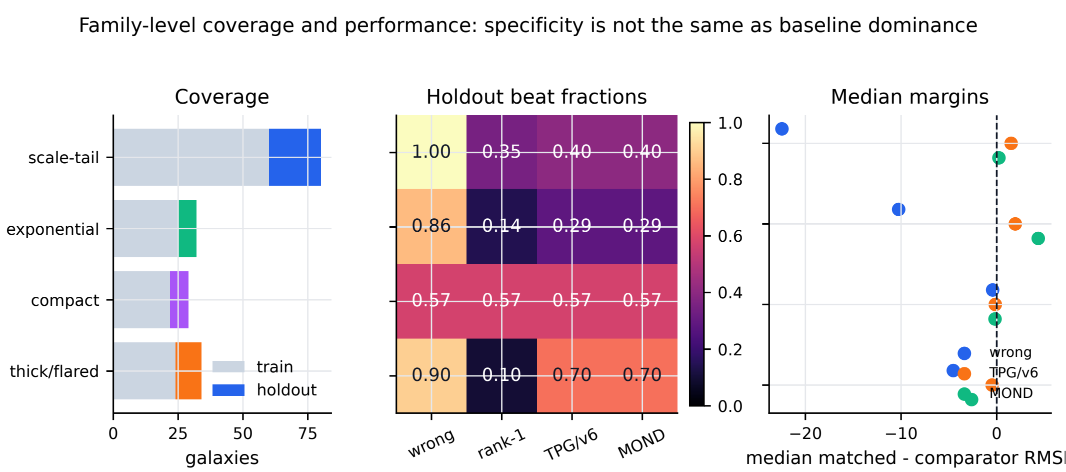

Figure 5: Family-level coverage and performance. Left: train/holdout coverage of the four one-dimensional source-native families. Middle: holdout beat fractions against wrong-family, rank-1, TPG/v6, and MOND-like comparators. Right: median matched-minus-comparator RMSE margins. The panel shows that coverage, morphology specificity, and proxy baseline flags are distinct quantities. Baseline-comparator beat fractions are proxy-level flags, not endpoint validation.

6.2 Inspected single-galaxy cases

The reproducibility code also contains four inspected cases. They are heterogeneous and should not be combined into a population validation claim.

Table 8: Four inspected readout cases. RMSE values are in km s^{-1}.

Galaxy	Matched	Best baseline	Best wrong	Status	Interpretation
NGC4013	10.61	10.88	11.37	retroactive/caveated	protocol-ready, replay needed
NGC5907	16.37	16.79	16.85	accepted preliminary	mixed projection readout
NGC7331	22.26	23.47	22.67	caveated accepted	mixed branch positive; pure branch negative
NGC4088	11.62	25.40	37.87	caveated accepted	strongest single case; law-level caveats

Across these four inspected rows, the source- or morphology-matched readout beats the best local baseline and the inspected wrong-family controls in all four cases. This is encouraging evidence for readout specificity. It is not population validation, and these cases were not selected as a statistically independent validation sample: NGC4013 remains retrospective, NGC7331 carries a broad outer-window caveat, and NGC4088 still has source-provenance and law-level normalization caveats. The object-level morphology context is residual-blind and source-ledger based: NGC4013 is supported by H I and vertical-structure studies [27, 3]; NGC5907 by warp, edge-on, and interaction/projection context [16, 18, 26]; NGC7331 by H I warp/trajectory and vertical-scale context [2, 14]. These references motivate the readout lanes and caveats; they are not by themselves endpoint evidence.

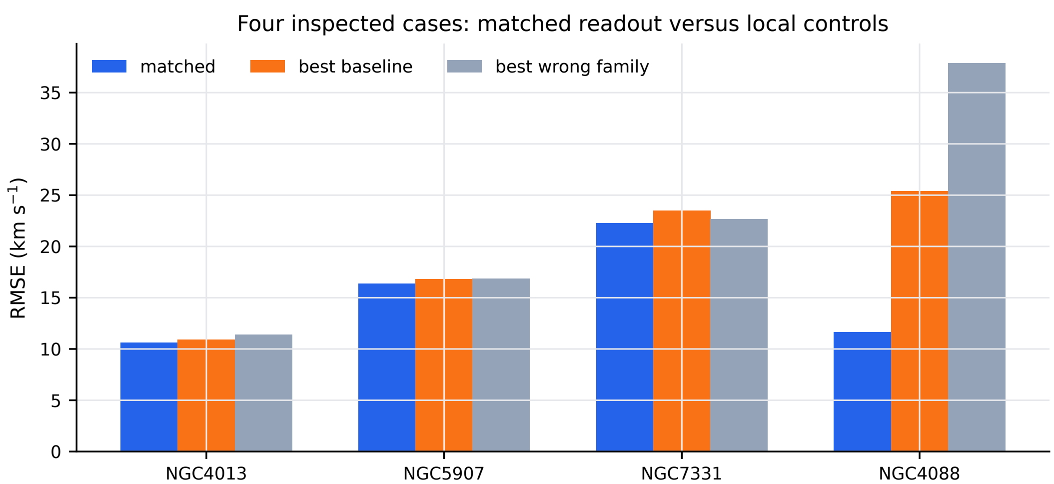

Figure 6: Four inspected cases. The matched readout is better than the best local baseline and best inspected wrong-family control in all four rows. The claim remains case-level and caveated because the evidence packets are heterogeneous.

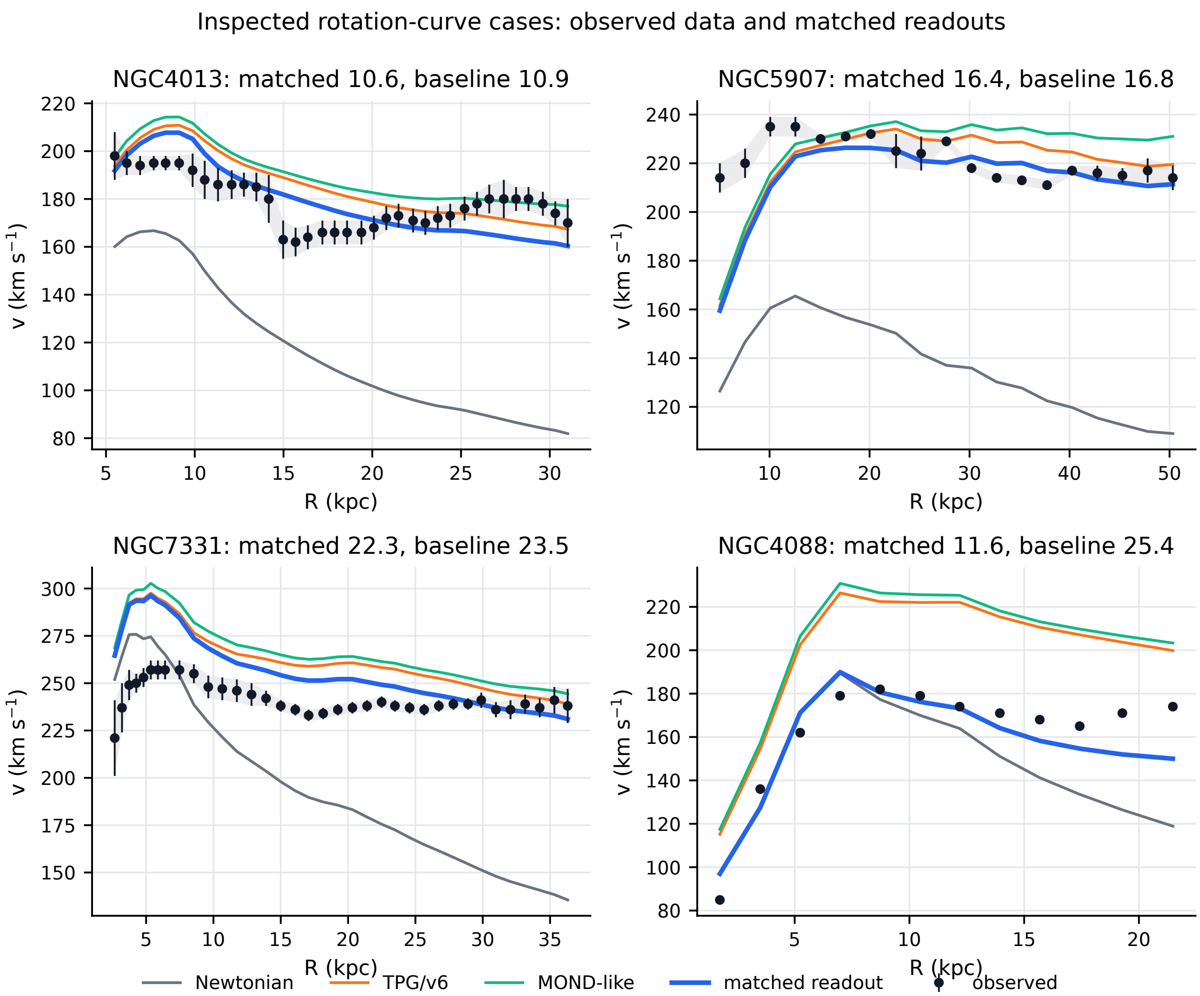

Figure 7: Rotation-curve view for the four inspected endpoint cases. Points are observed SPARC rotation speeds with reported error bands. The matched readout curve is compared with the Newtonian baryonic curve, the fixed TPG/v6 predecessor comparator, and the MOND-like comparator available in the reproducibility package. Panel titles report matched and best-baseline RMSE values.

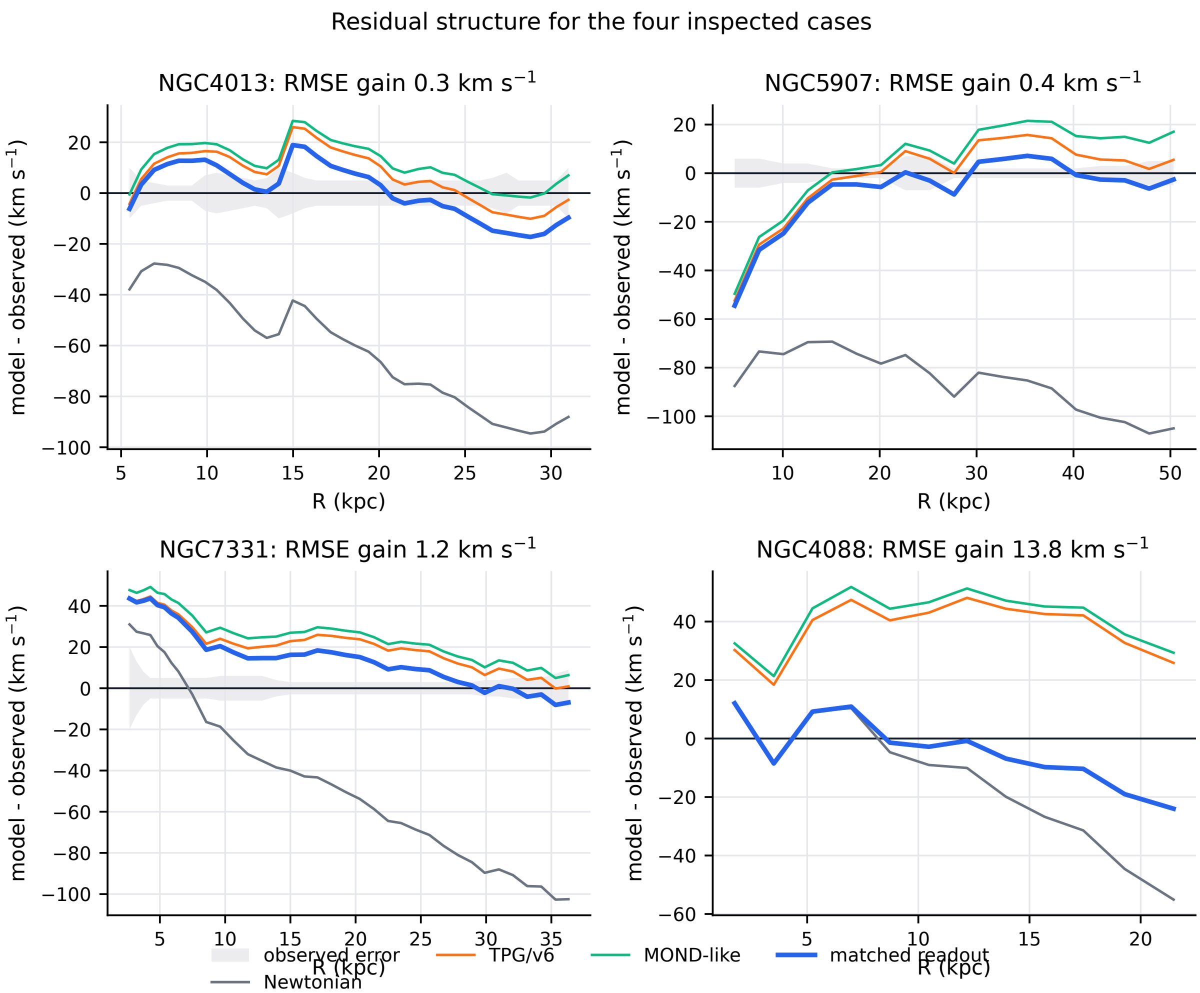

Figure 8: Signed residuals for the same four inspected cases, $\Delta v = v_{\text{model}} - v_{\text{obs}}$, with the observational error band shown around zero where available. Residual panels are important because an endpoint can look visually reasonable while leaving systematic radial structure. Panel titles report the matched improvement over the best local baseline in $\mathrm{km\,s^{-1}}$.

These rotation and residual panels are the most direct visual summary of the present single-galaxy evidence. NGC4088 shows the clearest endpoint-style gain: the matched warp/trajectory readout reduces a large baseline mismatch over much of the measured radial range. NGC5907 and NGC7331 are more modest: the matched mixed readout improves the score, but the curves remain close enough to strong smooth controls that the result should be read as a source-frozen case-improvement, not as a new law. NGC4013 is scientifically useful for a different reason: it shows how a mixed vertical/warp readout can become competitive, while also illustrating why retrospective cases need replay before promotion.

The TPG/v6 curves in these panels are therefore not a disposable comparison. They are a serious predecessor-control: when they are already close to the observations, a new morphology-matched readout must explain why it adds source-specific information rather than merely reproducing a smooth closure shape. This is why the paper separates three claims: morphology specificity against wrong families, competition against TPG/v6 or MOND-like controls, and full endpoint-grade validation.

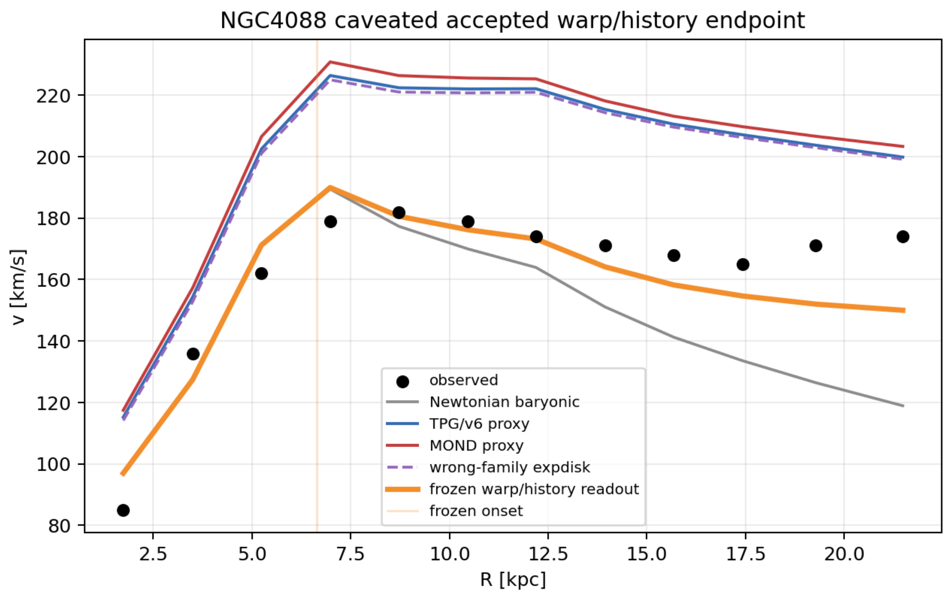

Figure 9: NGC4088 caveated warp/trajectory endpoint. This is the strongest single-galaxy result in the current code, but remains a preliminary control endpoint, not population validation.

An object-specific freeze note, `reports/ngc4088_object_specific_freeze_note.md`, records the residual-blind source fields, current blockers, and claim boundary for this case. The note is included because NGC4088 is visually compelling but should not be allowed to carry a population-level claim by itself.

7 Negative-result taxonomy

A poor score is not automatically a falsification of the morphology-readout framework. It must first be classified by source status.

How to read a failure. A failed branch says that one formula family or one component is insufficient. A weak negative says that the current morphology/source manifest is not strong enough to decide. A true negative requires a source-complete morphology manifest, a residual-blind frozen readout family and amplitude rule, endpoint scoring without retuning, and failure against relevant baselines plus wrong-family and shuffled-label controls.

1. *Branch-level negative control*: a particular readout branch fails, but another independently frozen branch may remain viable. The pure Φ_{HHC}-style interval branch for NGC7331 is an example; the mixed readout remains a separate caveated endpoint lane.

2. *Morphology-underdetermined weak negative*: the source evidence does not uniquely determine the readout family, mixture weights, projection state, or history proxy.

3. *Misclassification diagnostic*: a weak score suggests that the initial readout proxy may be wrong. This may motivate source acquisition or holdout design, but cannot promote a new family by itself.

4. *True negative candidate*: the morphology manifest is source-complete, the readout family and amplitude policy are frozen residual-blind, the endpoint is scored without retuning, and the result still fails against the relevant baselines and wrong-family controls.

At present the analysis contains branch-level negative controls and morphology-underdetermined weak negatives, but no source-complete endpoint-grade true negative against the framework.

This is not intended to make the framework unfalsifiable. It defines the minimum evidentiary standard under which a failed galaxy becomes an accepted true negative rather than a data-quality or label-quality diagnosis. In the next residual-blind catalogue campaign, a failed case will be promoted to an accepted true negative if all conditions in Table 9 are met before endpoint scoring.

Table 9: Predeclared future criteria for accepting a true negative against the morphology-readout framework.

Gate	Requirement	Failure interpretation
Source completeness	Independent morphology, H I/projection, vertical, support, and history fields are sufficient to select a readout-relevant family and subfamily.	If absent, the case is morphology-underdetermined.
Residual-blind freeze	The family, source scales, mixture weights if any, and amplitude policy are written to a freeze artifact before v_{obs}-residual scoring.	If absent, the case is diagnostic only.
Endpoint execution	The frozen readout is scored without retuning on the predeclared galaxy or holdout set.	If absent, no validation or falsification claim.
Control failure	The frozen readout fails against the relevant baseline set and does not beat wrong-family or shuffled-label controls.	If present with prior gates passed, accepted true negative.
Audit trail	The source ledger, scripts, random seeds, and output artifacts are preserved.	If absent, result is not reproducible enough to classify.

The current concrete negative and weak-negative examples are as follows.

- *Branch-level negative controls.* The pure Φ_{HHC}-style branch for NGC7331 is negative, but the mixed readout for the same galaxy improves the endpoint score. UGC08699 is best read as a compact-support branch control rather than as a general failure of the readout-morphology framework. Several weak exponential-disk population cases also behave as branch or proxy controls.

- *Morphology-underdetermined weak negatives.* NGC0247, NGC0300, NGC3917, NGC6015, NGC6503, UGC08286, and partly NGC3198 are not promoted to framework-failure cases in the present manuscript. For these objects the manifest may not yet determine the readout-relevant subfamily, projection or history component, mixture weight, or source-side amplitude strongly enough.

- *No accepted true negative yet.* A true negative would require source-complete morphology, a residual-blind frozen readout family and amplitude policy, endpoint scoring without retuning, failure against the relevant baselines, and wrong-family/shuffled controls that do not explain the failure. No case currently satisfies all of those requirements.

In plain operational terms, there are keys that do not open a given door, but there is not yet a source-complete case where the protocol shows that the correct key was used and the door still did not open.

8 Why a readout morphology catalogue is needed

The main bottleneck is not simply more curve fitting. The readout state of a galaxy requires information that is scattered across heterogeneous data products: NED classifications [13], S4G decompositions [19], SPARC mass models [8], DustPedia or other multiwavelength context [4], H I maps and radial profiles, PHANGS-like resolved gas or velocity-field products where available [9], and object-specific literature. Existing surveys contain many ingredients, but they were not designed to freeze morphology/readout states before rotation-curve endpoint scoring. For example, existing H I and edge-on studies already support pieces of such a catalogue for NGC3198 [1, 7, 17] and for several exploratory disturbed or dwarf-galaxy lanes [5, 25, 15, 21].

This suggests a concrete data product: a residual-blind readout morphology catalogue. Each row should record at least

$$\{K_{\rm obs}, K_{\rm readout}, \text{confidence tier, provenance, source windows, blocked fields}\},$$

plus disk scales, decomposition fields, H I support and tail/onset observables, vertical/projection flags, bar/ring/lopsidedness indicators, environment or morphology-trajectory evidence, and the allowed endpoint status. The catalogue should state whether a galaxy is endpoint-ready, replay-only, candidate-only, or morphology-underdetermined.

The morphology-trajectory entry is not an extra adjustable force. It records the possibility that the present projected morphology is not the whole readout state. A galaxy can currently look like a regular disk while carrying H I warp, interaction, outer-tail, vertical-support, settling, or relaxation evidence associated with a broader Tau-side morphology trajectory. In the protocol this information is allowed only as residual-blind source evidence; it cannot be inferred from the endpoint rotation residual after the fact.

The same caveat applies to observer/projection effects. They are not an after-the-fact permission to reinterpret any failed curve. They are catalogue fields to be frozen before scoring: inclination and line-of-sight geometry, resolved H I support, vertical/projection indicators, wavelength-dependent structure, and trajectory/phase flags. Only after those fields are frozen may a galaxy be assigned to a projected/readout-relevant subfamily.

Such a catalogue is observationally realistic. A pilot could begin with roughly 20–30 source-rich galaxies having SPARC rotation curves, good H I maps or velocity fields, infrared/optical decompositions, and usable multiwavelength or literature context. The readout state would be frozen first; only then would matched-family, wrong-family, shuffled-label, and baseline endpoint scores be run. This would replace ad hoc per-galaxy data hunting with a reproducible observing and reprocessing program.

Targeted observations could make the catalogue much stronger than the present patchwork. Existing surveys were not optimized to freeze readout morphology for this protocol, so the missing fields are unsurprising. A focused campaign would prioritize H I radial profiles and velocity fields, outer warp/tail onset radii, vertical scale or flare information for edge-on systems, bar/ring/lopsidedness indicators, and multiwavelength decomposition. Those are standard astronomical observables; the new element is the residual-blind combination of them into endpoint-ready readout states.

9 Discussion

The present results have two different meanings. First, morphology-indexed readout families appear to carry information: matched formulas can beat wrong families much more often than shuffled labels would suggest. Second, the present formulas are not yet a superior galaxy-rotation law. Their baseline performance is mixed, and the strongest single-galaxy results are caveated.

This is exactly why the endpoint protocol is useful. It prevents a weak result from being silently retuned into a fit, and it prevents a positive single-galaxy result from being overpromoted into population validation. The framework becomes scientifically interesting only when a source-complete readout catalogue allows predeclared morphology-family tests on a broader sample.

9.1 Why this is not yet a curve-fitting result

The present protocol was designed because unrestricted per-galaxy fitting would be scientifically uninformative. A sufficiently flexible radial function can usually be made to follow a rotation curve. The question asked here is different: can residual-blind source morphology/readout evidence choose a restricted readout family before the endpoint residuals are inspected? For that reason the paper separates formula construction, source-field acquisition, amplitude freezing, endpoint scoring, wrong-family controls, and shuffled-label controls.

Some of the current readout formulas are still idealized. They are morphology-indexed 4D readout proxies for thin disks, thick/flared disks, compact support, rings, warps, and mixed projection/trajectory lanes. A real galaxy can occupy a readout state that is richer than its present projected shape: interaction history, future-directed relaxation or settling phase, H I extent, vertical structure, bar/ring phase, lopsidedness, and inclination can all affect the effective readout. Therefore a weak result is not automatically evidence that the framework has failed. It may instead mean that the current formula is too coarse, the readout-relevant subfamily is wrong, or the source fields needed to freeze the right subfamily are missing. Only source-complete endpoint-grade failures are eligible to become true negatives.

9.2 Outlook: projection-enriched readout kernels

The present endpoint kernels should be read as morphology-indexed 1D readout proxies, not as complete observer/path Tau Core readouts. A natural next step is to replace a single morphology kernel by a residual-blind component sum with source-frozen observer/projection visibility weights,

$$\delta v^2_{\mathrm{proj}}(R) = \sum_j A_j(\mathrm{source})\, w_j(R, \mathcal{O}_{\mathrm{obs/path}}, \Theta_{\mathrm{morph}})\, K_j(R; K_{\mathrm{present}}, \Theta_{\mathrm{morph}}).$$

Here $\mathcal{O}_{\mathrm{obs/path}}$ denotes source-observer geometry such as inclination, position angle, line-of-sight projection, finite resolution, waveband, and velocity-field coverage, while Θ_{morph} denotes a source-frozen morphology trajectory/phase profile. The existing curves should therefore be read as present-only, observer/projection, or partial trajectory/phase approximations rather than full path-aware kernels. A still fuller path-aware layer would add a correction $\delta v^2_{\mathrm{path}}(R; \mathcal{P})$, where $\mathcal{P}$ is the source-observer null-geodesic bundle and its causal metric environment. That path-environment term is not modeled or scored in this paper.

Five inspected systems – UGC12506, NGC4013, NGC5907, NGC7331, and NGC4088 – suggest that such projection-enriched kernels are worth testing. They are not part of the headline population endpoint here. They define the next predeclared catalogue test: source-frozen observer/projection labels should select the enriched kernel family before endpoint scoring, and the matched enriched families should be compared against simpler morphology-only proxies, wrong enriched families, shuffled observer/projection labels, and conventional baselines. The detailed derivation, candidate audit, and validation gate are therefore separated into the companion projection-enriched readout paper.

9.3 How conventional baselines fit into the readout picture

The paper also does not assume that Newtonian, MOND/RAR-like, or TPG/v6-like curves should fail everywhere. A regular, stable, low-disturbance disk can be close to a Newtonian or smooth scalar readout. A galaxy whose residuals are well described by a single acceleration scale can be naturally strong under a MOND/RAR-like comparator. A galaxy dominated by a smooth closure response can be strong under the frozen TPG/v6 predecessor comparator. These are not contradictions of the morphology-readout programme; they are limiting cases the programme must explain.

The stronger claim would be this: after source-complete morphology labels are frozen, the correct readout subfamily should explain when those simple limits are adequate and when a richer morphology-specific kernel is needed. The present paper does not prove that claim. It sets up the endpoint machinery needed to test it.

10 Conclusion

This paper reframes morphology-sensitive rotation-curve work as a forward readout problem. The central claim is not that a new gravity theory has been validated, nor that every galaxy can be rescued by a custom curve. The claim is that a residual-blind morphology/readout state can be used to choose a predeclared readout family, and that this choice can be tested against wrong families, shuffled labels, and conventional baselines.

The current numerical evidence is preliminary but useful. The strongest broad signal is morphology specificity: source-native readout formulas beat wrong formula families in 0.886 of 44 holdout galaxies with a shuffled-label $p \simeq 0.002$ for the mean matched-minus-wrong statistic; the beat-fraction null is reported separately. The same formulas do not yet beat TPG/v6 or MOND-like comparators as a population. Four inspected galaxies are retained here only as caveated demonstrations of the endpoint workflow. The more detailed observer/projection-enriched kernel derivation and its four-object audit are separated into a companion paper, because they define a new validation programme rather than the headline population endpoint of the present paper.

The next step is therefore not another round of residual-driven curve repair. It is a readout morphology catalogue and a predeclared endpoint campaign. If the correct source-frozen readout family systematically beats wrong families and shuffled labels, and then becomes competitive against Newtonian, MOND/RAR, and TPG-like baselines, the framework will have a meaningful empirical target. If source-complete endpoint-grade cases fail, those failures should be retained as true negative candidates. The absence of such a case in the present analysis is a limitation of the current source catalogue, not a positive result.

A Readout-family derivation ledger

This appendix records the manuscript-level derivations of the readout kernels used in the present population preflight. The purpose is not to prove a final gravity law. It is to make explicit which part is a definition, which part is a source ansatz, which part follows from a standard radial readout/closure calculation, and which part remains an empirical or source-completeness obligation.

This appendix is theory-motivation scaffolding. The numerical test in the main text does not require the reader to accept the full Tau Core bridge ontology. Operationally, the test only requires residual-blind labels, frozen source proxies, predeclared readout families, and matched-versus-wrong/shuffled controls. The bridge equations below explain why these families are grouped as readout specializations, but the population statistics can be audited from the saved artifacts even if the parent theory remains unaccepted.

Dependency boundary: the endpoint statistics do not depend on accepting Appendix A. Appendix A only motivates the grouping of the readout families and records the formula ledger used by the reproducibility package.

A.1 Tau Core bridge equation used by the readout families

The readout families in this manuscript are motivated as reductions of one common conditional bridge object. Let P_D^τ denote the Tau-side ordered-disk readout slice and let P_N^τ denote the Newtonian/metric closure-test slice. Their residual-blind local mismatch is

$$\delta P_\tau = P_N^\tau - P_D^\tau.$$

Only the tangent/off-slice part of this mismatch can generate a readout residual. Denoting that part by $\delta P_\tau^\perp$, the first-order relative generator is

$$\mathcal{M}_\tau = [\delta P_\tau^\perp, P_D^\tau].$$

The bridge then tests whether this generator moves the ordinary diagonal disk sector outside the Newtonian, ordinary-disk, and gauge subspace

$$B_{\mathrm{axis}} = N + D_{\mathrm{diag}} + G.$$

The non-absorption map is

$$\rho_D(\mathcal{M}_\tau) := \pi_B \circ \mathcal{M}_\tau|_{D_{\mathrm{diag}}}, \qquad \pi_B : V_{\mathrm{grav}}^\tau \to V_{\mathrm{grav}}^\tau / B_{\mathrm{axis}}.$$

The common 4D weak-field readout equation used by the present paper is therefore

$$\boxed{\delta g_{\mathrm{morph}}(x) = R_{g,x}^{4D}\left[\rho_D(\mathcal{M}_\tau(x))(Q_D^\tau(x))\right].} \tag{1}$$

Here Q_D^τ is the morphology-selected ordered-disk trace-free direction. Equation (1) is the bridge base equation used in this paper. It is not a final law of gravity; it is a conditional readout rule: if the Tau-side slice mismatch is source-defined, survives the quotient, and has a nonzero 4D weak-field readout, then it contributes an effective morphological residual.

For the local disk witness used throughout the bridge, let S_D^τ be the in-disk shear-like partner of Q_D^τ. If the closure-test slice selects

$$Q_N^\tau = \cos\alpha_\tau\, Q_D^\tau + \sin\alpha_\tau\, S_D^\tau,$$

then, to first order in the local mismatch angle,

$$\mathcal{M}_\tau = \alpha_\tau J_D, \qquad J_D(Q_D^\tau) = S_D^\tau, \qquad J_D(S_D^\tau) = -Q_D^\tau.$$

Consequently

$$\rho_D(\mathcal{M}_\tau)(Q_D^\tau) = \alpha_\tau [S_D^\tau]. \tag{2}$$

Equation (2) is the algebraic bridge reduction. It says that the residual is not produced by inserting a new force; it is the quotient-visible part of a readout/closure mismatch. The residual vanishes if $\alpha_\tau = 0$, if S_D^τ is absorbed into B_{axis}, or if the 4D readout map kills this quotient class.

A.2 From bridge residual to a radial family kernel

To compare with one-dimensional rotation curves, the bridge readout must be reduced to a radial observable. In the present preparation-level tests the family label K chooses a residual-blind source functional $\mathcal{S}_K^\tau$ representing the observable part of $\alpha_\tau[S_D^\tau]$ for that morphology/readout class:

$$\alpha_\tau[S_D^\tau] \quad \longrightarrow \quad \mathcal{S}_K^\tau.$$

The 4D readout map then induces a radial operator $\mathcal{R}_K$,

$$\delta g_{K,R}(R) = \mathcal{R}_K[\mathcal{S}_K^\tau](R),$$

and circular-orbit comparison uses

$$\delta v_K^2(R) = R\,\delta g_{K,R}(R). \tag{3}$$

Thus every endpoint shell below has the common form

$$\delta v_K^2(R) = A_K K_K(R),$$

where $K_K(R)$ is the source-selected radial kernel and A_K is the allowed source amplitude or frozen amplitude rule.

This is the derivation status used in the paper:

- The slice-mismatch and quotient equations (1)–(2) are the conditional Tau Core bridge scaffold.
- The choice of $\mathcal{S}_K^\tau$ is a residual-blind morphology/readout source ansatz, not an endpoint fit.
- The radial kernels $K_K(R)$ follow from applying the chosen radial readout operator $\mathcal{R}_K$ to that source ansatz.
- The current population preflight does not prove the parent Tau Core theory or the exact amplitudes A_K. It tests whether the residual-blind family specialization carries information by comparing matched families, wrong families, shuffled labels, and baseline curves.

Table 10: Logical status of the bridge-to-kernel construction.

Layer	Equation or object	Status in this manuscript
Tau-side slice mismatch	$\delta P_\tau = P_N^\tau - P_D^\tau$, $\mathcal{M}_\tau = [\delta P_\tau^\perp, P_D^\tau]$	Definition within the conditional bridge scaffold.
Quotient non-absorption	$\rho_D = \pi_B \circ \mathcal{M}_\tau\vert_{D_{\mathrm{diag}}}$, $\rho_D(\mathcal{M}_\tau)(Q_D^\tau) = \alpha_\tau[S_D^\tau]$	Conditional algebraic bridge result.
4D readout shell	$\delta g_{\mathrm{morph}} = R_g^{4D}[\rho_D(\mathcal{M}_\tau)(Q_D^\tau)]$	Bridge base equation tested operationally here.
Family source specialization	$\alpha_\tau[S_D^\tau] \to \mathcal{S}_K^\tau$	Residual-blind source ansatz, family by family.
Radial endpoint kernel	$\delta v_K^2 = R\,\mathcal{R}_K[\mathcal{S}_K^\tau] = A_K K_K(R)$	Derived shell after choosing the source class and radial readout operator.
Amplitude and source completeness	A_K, source radii, thickness, support, projection state	Frozen, train-split, or proxy-level depending on the endpoint row.

A.3 Common normalization and dimensional rule

For every family K, the executable preflight uses a velocity-squared readout of the form

$$v_{gK}^2(R) = v_{\rm base}^2(R) + \delta v_{gK}^2(R), \qquad \delta v_{gK}^2(R) = A_{gK} K_{gK}(R).$$

Here $K_{gK}(R)$ is the source-selected radial kernel and A_{gK} is the corresponding amplitude. The kernel need not be dimensionless. The dimensional requirement is simply

$$[A_{gK}]\,[K_{gK}] = (\text{velocity})^2.$$

In a strict endpoint, A_{gK} must be source-derived or frozen by a predeclared rule. In the current population preflight, the family shapes are the derived readout shells, while several source scales and amplitudes remain available-data proxies or train-split normalizations. That is why the population result is evidence for morphology specificity, not a final empirical validation.

A.4 Scale-tail or diffuse LSB readout

Definition/source ansatz. The scale-tail family is used when the source morphology indicates an extended outer load, diffuse low-surface-brightness support, or an H I/tail-like readout component. The residual-blind source inputs are an inner onset radius $R_{\rm in}$ and an outer cutoff/support radius $R_{\rm cut}$. The minimal $n = 2$ tail window is

$$W_{\rm tail}(r) = \frac{(r/R_{\rm in})^2}{1 + (r/R_{\rm in})^2}\frac{1}{1 + (r/R_{\rm cut})^2}.$$

The first factor suppresses the inner disk; the second keeps the source finite.

Bridge specialization. In Eq. (1), the scale-tail family corresponds to a source class $\mathcal{S}^{\tau}_{\rm tail} = A_{\rm tail} W_{\rm tail}(r)$ whose quotient-visible readout is dominated by an extended outer support. The radial operator $\mathcal{R}_{\rm tail}$ is the one-dimensional accumulated-load readout. Thus the family is obtained by applying $\delta v^2 = R\,\mathcal{R}_{\rm tail}[\mathcal{S}^{\tau}_{\rm tail}]$, not by fitting an arbitrary tail shape to the rotation residual.

Radial readout. For a one-dimensional preflight, the allowed radial load is the accumulated tail source

$$I_2(R) = \int_0^R W_{\rm tail}(r)\,dr.$$

The corresponding scale-tail velocity-squared response is taken to be

$$\boxed{\delta v_{\rm tail}^2(R) = A_{\rm tail}\frac{I_2(R)}{R}} \qquad K_{\rm tail}(R) = I_2(R)/R.$$

This is the formula implemented in the reproducibility code as the $n = 2$ scale-tail shell.

Limits. For $R \ll R_{\rm in}$, $W_{\rm tail} \sim (R/R_{\rm in})^2$, so the tail readout is suppressed in the inner disk. For $R_{\rm in} \ll R \ll R_{\rm cut}$, $I_2(R) \sim R$, so $I_2(R)/R$ approaches a slowly varying or flat contribution. For $R \gg R_{\rm cut}$, $I_2(R)$ saturates and the response falls as $1/R$ in velocity squared. Thus the shell has the intended inner-suppressed, outer-supported, finite-tail behavior without reading endpoint residuals.

Claim boundary. This derivation fixes the readout shape once R_{in} and R_{cut} are source-frozen. It does not by itself derive the amplitude A_{tail}, nor does it prove that every diffuse or LSB galaxy belongs to this family.

A.5 Regular exponential-disk readout

Definition/source ansatz. The exponential-disk family is used when the source morphology is a regular axisymmetric disk with a residual-blind scale radius R_s. The source surface readout is

$$\Sigma_{\text{exp}}(R) = \Sigma_0 e^{-R/R_s}.$$

Bridge specialization. In the bridge notation, this family chooses $\mathcal{S}^{\tau}_{\text{exp}} = A_{\text{exp}}\Sigma_0 e^{-R/R_s}$ as a regular axisymmetric source class. The corresponding radial operator $\mathcal{R}_{\text{exp}}$ is the standard axisymmetric disk Green operator. It is included as a conservative bridge specialization: the Tau-side family selection decides when the regular-disk carrier is the appropriate readout, while the carrier itself is the familiar exponential-disk radial response.

Radial readout. The axisymmetric thin-disk Green-function calculation for an exponential disk gives the standard Freeman radial carrier [6]

$$v^2_{\text{exp}}(R) \propto y^2 \left[I_0(y)K_0(y) - I_1(y)K_1(y)\right], \qquad y = \frac{R}{2R_s},$$

where I_n and K_n are modified Bessel functions. The preflight kernel keeps this source-fixed shape and absorbs dimensional constants into the amplitude:

$$\boxed{\delta v^2_{\text{exp}}(R) = A_{\text{exp}}\, R_s\, y^2 \left[I_0(y)K_0(y) - I_1(y)K_1(y)\right]}$$

or

$$K_{\text{exp}}(R) = R_s\, y^2 \left[I_0(y)K_0(y) - I_1(y)K_1(y)\right].$$

Limits. The response is finite at the origin, peaks on disk scales, and declines at large radii. It is therefore a regular-disk carrier, not an arbitrary long-range tail.

Claim boundary. This is the most conservative readout family: it imports the standard axisymmetric exponential-disk radial carrier as the morphology-matched shell. The current paper does not claim that this is a new gravity formula. It tests whether source morphology can decide when this restricted shell is the right readout family.

A.6 Compact finite-source readout

Definition/source ansatz. The compact family is used when the source morphology indicates centrally concentrated or finite support with an outer compact radius R_c. The minimal assumption is that the active readout load is finite:

$$I_{\text{compact}}(R) \longrightarrow I_\infty \qquad (R > R_c).$$

Bridge specialization. Here $\mathcal{S}^{\tau}_{\text{compact}}$ is the finite-support quotient-visible source class produced by Eq. (1). The radial readout operator $\mathcal{R}_{\text{compact}}$ is used only in its exterior finite-load limit unless stronger source information is available. Therefore the $1/R$ velocity-squared shell is a compact-source consequence, while the inner continuation used by the code is a numerical regularization.

Exterior readout. Outside the support, any finite radial load has the leading exterior behavior

$$\delta g_R(R) \sim -\frac{B_c}{R^2},$$

and therefore

$$\boxed{\delta v^2_{\mathrm{compact}}(R) = R\,|\delta g_R(R)| \sim \frac{B_c}{R}} \qquad (R \geq R_c).$$

The executable preflight uses

$$K_{\mathrm{compact}}(R) = \begin{cases} R^2/R_c^3, & R < R_c, \\ 1/R, & R \geq R_c, \end{cases}$$

so that the kernel is finite in the inner numerical domain and matches the exterior $1/R$ value at $R = R_c$.

Limits. The scientifically meaningful part of this shell is the exterior finite-load falloff. The inner regularization is a numerical preflight convention, not a claim about the detailed central dynamics.

Claim boundary. This family is endpoint-ready only when R_c is source-native or frozen residual-blind. A compact label without a compact-support source field is a candidate or control, not a validation row.

A.7 Thick/flared or projection-sensitive readout

Definition/source ansatz. The thick/flared family is used when the source morphology indicates vertical support, flaring, edge-on projection, or projection-sensitive readout. The source inputs are a disk scale R_s and a dimensionless thickness profile $h(R)/R_s$, or a frozen constant proxy when no radial profile is available.

Bridge specialization. For this family, Eq. (1) is reduced with a three-dimensional/projection-sensitive source class $\mathcal{S}^{\tau}_{\mathrm{flare}}(R, z)$. In a one-dimensional SPARC preflight, the vertical part cannot be fully resolved, so its effect enters the radial operator $\mathcal{R}_{\mathrm{flare}}$ as a damping of short-wavelength Hankel modes. This is why the shell is a projection/readout attenuation, not a generic extra force.

Damped Fourier-Bessel readout. The thin exponential-disk response can be written as a Hankel/Fourier-Bessel radial integral. Vertical thickness damps high spatial-frequency modes. The minimal constant-h damping used in the preflight is

$$D(u, h/R_s) = \frac{1}{1 + u\,h/R_s}.$$

With $x = R/R_s$, the thick/flared shell is

$$\boxed{K_{\mathrm{flare}}(R) = R_s\,\frac{x}{2}\int_0^{\infty} \frac{u\,J_1(ux)}{(1+u^2)^{3/2}}\,\frac{du}{1 + u\,h/R_s}}$$

and

$$\delta v^2_{\mathrm{flare}}(R) = A_{\mathrm{flare}} K_{\mathrm{flare}}(R).$$

Limits. When $h/R_s \to 0$, the damping tends to unity and the shell reduces to the thin-disk radial carrier class. Larger h/R_s suppresses short-wavelength radial structure and gives a smoother projected response. This is the desired behavior for vertical/projection-sensitive systems.

Claim boundary. The current SPARC population run mostly uses scalar thickness proxies, so this family is a one-dimensional proxy for a genuinely three-dimensional readout. It becomes endpoint-grade only when vertical/projection observables are source-frozen strongly enough.

A.8 Ring, barred, and lopsided families

The registry also contains ring/resonance, barred $m = 2$, and lopsided $m = 1$ families. Their source forms are

$$\Sigma_{\rm ring}(R) \propto \exp\left[-\frac{(R - R_{\rm ring})^2}{2w_{\rm ring}^2}\right],$$

and

$$\Sigma_m(R, \phi) = \Sigma_m(R) \cos[m(\phi - \phi_m)], \qquad m = 1, 2.$$

These are legitimate morphology handles, but they are not promoted here to full one-dimensional endpoint kernels. The ring family can sometimes be used as a localized radial proxy if $R_{\rm ring}$ and $w_{\rm ring}$ are source-frozen. The $m = 1$ and $m = 2$ families are intrinsically velocity-field objects because their phase information is largely lost in a one-dimensional azimuthally averaged rotation curve. Treating them as full SPARC-1D endpoint families would overstate the evidence.

A.9 Summary of derivation status

Table 11: Readout-family derivation status. "Derived shell" means that the kernel shape follows from the stated source ansatz and radial readout calculation. It does not mean that the amplitude or source classification is empirically validated.

Family	Derived kernel	Required source freeze	Status in this paper
$K_{\rm scale\_tail\_spiral}$	$K_{\rm tail} = I_2(R)/R$, $I_2 = \int_0^R W_{\rm tail}(r)dr$	$R_{\rm in}$, $R_{\rm cut}$, amplitude policy	Derived 1D shell; source scales often proxy.
$K_{\rm exponential\_disk}$	$K_{\rm exp} = R_s y^2[I_0K_0 - I_1K_1]$	R_s, amplitude policy	Derived standard disk carrier; conservative 1D shell.
$K_{\rm compact\_finite}$	$K_{\rm compact} \sim 1/R$ outside R_c	R_c, finite-load evidence, amplitude policy	Exterior shell derived; inner regularizer is numerical.
$K_{\rm thick\_flared}$	damped Fourier-Bessel kernel with $D = (1 + uh/R_s)^{-1}$	R_s, h/R_s or profile, projection state	Derived 1D proxy for 3D/projection readout.
$K_{\rm ring}$	localized radial source convolution near $R_{\rm ring}$	$R_{\rm ring}$, $w_{\rm ring}$, phase/context	Proxy only in this paper.
$K_{m=1}, K_{m=2}$	harmonic velocity-field response	$\Sigma_m(R)$, phase ϕ_m, velocity field	Velocity-field preferred; not a full 1D endpoint kernel.

B Holdout per-galaxy freeze ledger

Table 12 is an abbreviated ledger for the 44 holdout galaxies behind the 0.886 source-native formula result. The full machine ledger is `data/derived/source_native_readout_formula_labels.`

`csv`; the score ledger is `data/derived/source_native_readout_formula_scores_by_galaxy.csv`. The source-proxy vector is reported as

$$R_s/R_{\rm in}/R_{\rm cut}/R_c/(h/R_s).$$

These are not claimed to be final source-native observables; they are the frozen available-data proxies used by the executable preflight. Caveat codes: FPTS = few rotation points, LINC = low inclination, LDE = large distance error, and VPROXY = vertical geometry proxy only. The final column reports matched-minus-wrong-family mean RMSE, so negative Δ means the matched family beats the wrong-family mean.

Table 12: Abbreviated holdout per-galaxy label/source/score ledger.

Galaxy	Family	Conf.	Role	Caveat	Source vector	Outcome
CamB	tail	1.00	middle	none	0.54/0.32/2.09/0.90/0.23	win; −71.52
ESO116-G012	exp	0.85	middle	LDE	2.46/1.45/10.95/4.13/0.18	win; −10.25
F561-1	tail	0.60	candidate	FPTS,LINC	3.36/1.98/10.89/5.64/0.19	win; −13.86
F563-1	tail	0.65	candidate	LINC,LDE	5.52/3.24/24.89/9.26/0.29	win; −2.90
F568-1	flare	0.70	candidate	LINC,VPROXY	3.09/1.81/14.68/5.19/0.18	loss; 5.40
IC2574	tail	1.00	control	none	3.30/1.94/12.16/5.54/0.25	win; −16.19
IC4202	compact	1.00	candidate	none	7.69/4.52/26.52/18.33/0.10	win; −0.43
NGC0024	flare	0.90	middle	VPROXY	1.01/0.59/11.84/1.70/0.13	win; −26.51
NGC2683	flare	0.90	middle	VPROXY	10.32/6.06/35.26/18.95/0.10	win; −2.99
NGC3726	flare	0.90	candidate	VPROXY	7.79/4.57/34.51/13.06/0.14	win; −4.36
NGC3949	flare	0.70	middle	FPTS,VPROXY	2.60/1.53/7.29/4.37/0.11	win; −14.09
NGC3972	flare	0.90	middle	VPROXY	2.86/1.68/9.07/4.80/0.12	win; −11.38
NGC4013	compact	1.00	middle	none	10.86/6.38/32.08/22.26/0.11	loss; 5.33
NGC4088	flare	0.90	candidate	VPROXY	6.75/3.97/22.44/11.33/0.12	win; −4.72
NGC4183	exp	1.00	control	none	6.24/3.66/24.28/10.47/0.22	win; −7.54
NGC4214	tail	0.80	middle	LINC	1.74/1.02/6.14/2.92/0.16	win; −23.65
NGC4389	flare	0.70	candidate	FPTS,VPROXY	1.87/1.10/5.38/3.15/0.09	win; −21.07
NGC5907	flare	0.90	middle	VPROXY	16.50/9.69/54.29/27.68/0.15	win; −1.22
NGC5985	compact	0.85	candidate	LDE	9.32/5.47/35.23/17.80/0.09	win; −7.42
NGC6789	tail	0.80	middle	FPTS	0.24/0.14/0.82/0.41/0.22	win; −107.71
NGC6946	compact	0.65	middle	LINC,LDE	6.13/3.60/21.14/11.86/0.11	loss; 13.68
NGC7331	flare	0.90	candidate	VPROXY	10.49/6.16/37.61/18.84/0.11	win; −1.84
UGC00731	tail	0.85	middle	LDE	3.52/2.07/14.72/5.91/0.39	win; −16.98
UGC00891	tail	0.65	control	FPTS,LDE	2.65/1.56/9.79/4.45/0.37	win; −23.48
UGC02023	tail	0.45	middle	FPTS,LINC,LDE	1.35/0.79/4.17/2.27/0.17	win; −37.05
UGC02259	exp	0.85	middle	LDE	2.73/1.60/9.43/4.58/0.22	win; −8.92
UGC03580	compact	0.85	candidate	LDE	1.64/0.96/28.70/4.07/0.13	loss; 28.48
UGC04305	tail	1.00	candidate	none	1.72/1.01/6.42/2.88/0.23	win; −28.17
UGC04499	tail	0.85	control	LDE	2.71/1.59/9.82/4.55/0.26	win; −18.61
UGC05414	tail	0.65	control	FPTS,LDE	1.43/0.84/4.78/2.40/0.23	win; −33.83
UGC05716	tail	0.85	control	LDE	4.00/2.35/16.03/6.71/0.35	win; −17.56
UGC05829	tail	0.65	control	LINC,LDE	2.25/1.32/8.32/3.77/0.26	win; −25.96

Continued on next page

Galaxy	Family	Conf.	Role	Caveat	Source vector	Outcome
UGC05986	tail	0.85	middle	LDE	2.99/1.76/10.17/5.02/0.15	win; −9.47
UGC06818	tail	1.00	middle	none	2.35/1.38/7.62/3.94/0.16	win; −21.42
UGC06917	tail	1.00	control	none	3.64/2.14/11.44/6.11/0.16	win; −14.68
UGC06983	tail	1.00	middle	none	5.20/3.05/18.45/8.72/0.24	win; −8.10
UGC07151	exp	1.00	control	none	1.79/1.05/6.04/3.00/0.17	win; −29.87
UGC07261	exp	0.45	control	FPTS,LINC,LDE	2.27/1.33/7.36/3.81/0.17	win; −22.60
UGC07577	tail	1.00	middle	none	0.56/0.33/1.78/0.94/0.13	win; −69.53
UGC07603	exp	0.85	middle	LDE	1.32/0.78/4.80/2.22/0.23	win; −32.16
UGC08699	compact	0.85	middle	LDE	4.27/2.51/26.33/10.75/0.10	win; −2.94
UGC11914	compact	0.65	candidate	LINC,LDE	2.28/1.34/9.85/6.01/0.08	win; −10.70
UGC12506	exp	1.00	candidate	none	15.14/8.89/55.56/25.41/0.18	loss; 4.99
UGCA444	tail	1.00	control	none	0.81/0.48/3.74/1.36/0.47	win; −54.28

C Source-reference ledger

Table 13 records how the main external references enter the present paper. The entries are provenance for residual-blind source fields and baseline context. They do not by themselves validate any readout formula.

Table 13: Reference ledger for source fields used or proposed by the morphology-readout protocol.

Reference group	Role in this paper	Claim boundary
[8]	SPARC rotation curves and baryonic mass-model arena.	Common numerical data source; not a readout-morphology catalogue.
[12, 11, 10]	MOND-like and empirical RAR comparison context.	Baseline comparators; not used to choose morphology-readout families.
[6]	Standard exponential-disk radial carrier used in the regular-disk readout derivation.	Formula-context reference; not evidence for the morphology-matched endpoint protocol by itself.
[22, 24, 20, 23]	Examples of H I, velocity-field, late-type photometric/kinematic, and lopsidedness data products suitable for readout morphology fields.	Support feasibility of a catalogue; object-specific endpoint gates still require per-galaxy source freeze.
[13, 19, 4, 9]	Catalogue, decomposition, multiwavelength, and resolved-gas infrastructure needed for a residual-blind readout morphology catalogue.	Source-infrastructure references; they support data availability and provenance, not endpoint validation by themselves.
[1, 7, 17]	NGC3198 H I extent, extraplanar gas, and radial-flow/tail-onset source context.	Used to illustrate source acquisition and failure-mode routing; not promoted as an accepted endpoint in this paper.
[27, 3]	NGC4013 H I and vertical-structure context.	Supports mixed/vertical-overlay interpretation; NGC4013 remains retrospective and replay-needed.
[16, 18, 26]	NGC5907 warp, edge-on, and projection/ISM context.	Supports the projection/mixed readout lane; single-galaxy result remains preliminary.
[2, 14]	NGC7331 H I warp/history and vertical-scale context.	Supports caveated mixed endpoint and pure-branch negative-control interpretation.
[24]	NGC4088 H I geometry, strong disk distortion, position–velocity asymmetry, asymmetric warp behaviour, and near-companion context.	Supports the warp/history/asymmetric-projection readout lane; current NGC4088 result remains source-review and normalization-law caveated.
[5, 25, 15, 21]	Exploratory disturbed/dwarf H I morphology source classes.	Included as examples for future catalogue lanes; not scored as endpoint evidence here.

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